Francesco Calice

+44 7740 776507 | francesco.calice23@imperial.ac.uk | Research Portfolio

Education

Imperial College London — London, UK

MSc Quantum Fields and Fundamental Forces (Incoming) — *Sep. 2026 – Aug. 2027*

BSc Physics with Theoretical Physics | 1st Class Honours — *Sep. 2023 – Jun. 2026*

- **84.98%** third year average - **Dean's List Prize** for top 10% performance.
- Ranked **2/235** in Physics Lab and Analysis Module.
- Relevant Modules: Group Theory (92%), Advanced Classical Physics (92%), Foundations of Quantum Mechanics (90%).

Eton College — Windsor, UK

*A-Levels: 4 A*s* — *Sep. 2018 – Jul. 2023*

- Awarded two academic scholarships: Honoris Causa (2018-2023), Oppidan Scholar (2019-2023).

Research Experience

CERN – Neutrino Group — Geneva, Switzerland

Summer Student | EP-Nu, Supervisor: Dr. Animesh Chatterjee — *Jun. 2026 – Aug. 2026*

- Selected for the **CERN Summer Student** programme from approximately 20,000 applicants; one of ~ 20 students from the UK.
- Performed a **beyond-standard-model phenomenology study** of heavy-neutral-lepton production and decay at ProtoDUNE-SP; developed a neutrino-background-rejection algorithm and derived new background-inclusive projected sensitivities to active-sterile mixing, U_{e4}^2 and $U_{\mu 4}^2$.
- Presented results at EP-Nu internal conferences and authored a summary report documenting the analysis, validation and projected sensitivity limits.

Imperial College London – Dark Matter Direct Detection — London, UK

Bachelor's Research Project | Supervisor: Prof. Stefan Söldner-Rembold — *Oct. 2025 – Jan. 2026*

- Modelled spin-independent WIMP-nucleon interactions in Python; **derived analytic cross-sections**.
- Developed a toy model for detector cross-section limits from first principles to inform detector design principles.
- Awarded top marks in year (**86%**)

Imperial College London – Type Ia Supernovae — London, UK

Undergraduate Researcher | Supervisor: Dr. Eleni Tsaprazi — *Jan. 2025 – Mar. 2025*

- Built a full SN Ia rate-modelling pipeline in **Python**, analysing $\sim$ 150M galaxies from **SQL** galaxy catalogues.
- Implemented A+B and integrated star-formation-history models; carried out Poisson sampling and rate–morphology analyses using SAG, SAGE, and GALACTICUS.
- Presented results at an international supernovae and cosmology **teleconference**; summary paper incorporated into subsequent grant proposal.

Independent Projects

Interactive Theorem Proving for AI-assisted Theoretical Physics — Aug. 2026 – Present

Independent Research and Working Paper — Lean 4, Physlib, SAT/PB

- Preparing a short paper on adapting **end-to-end formal verification** from mathematics to physics: from axiomatising physical assumptions and encoding theories in Lean4 and Physlib, through machine-checked derivations, to AI-assisted construction and comparison of candidate theories.
- Investigating the **Krenn–Gu conjecture** on monochromatic quantum graphs as a case study, combining symmetry reductions, exact SAT/PB searches and machine-checkable Lean proof components, with an initial focus on the $N = 8, D = 3$ subcase.

Machine Learning in Chaotic Systems — Apr. 2024 – Jul. 2024

Imperial College London (Summer Project)

- Implemented and compared recurrent architectures, including vanilla **RNNs and LSTMs**, for multi-step forecasting of simulated chaotic triple-pendulum trajectories.
- Built the data-processing, training and evaluation workflow and ran model experiments on Imperial's **HPC GPU** nodes.
- Investigated the challenges posed by nonlinear dynamics and sensitivity to initial conditions; presented the methodology and results internally.

Hobbies and Interests

- Rowed for Imperial College at **Henley Royal Regatta** in the Prince Albert Challenge Cup
- Demonstrator in technical skiing videos used for **Austrian National Skiing** instructor program
- Fluent in **English**, **German** and **Italian**